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| Document number | PLS-SDS-MP-HIV-001                                                                                     |
| Revision        | Approved v1.0                                                                                          |
| Issue date      | 09 October 2026                                                                                        |
| Product format  | Lyophilized PCR tube-strip microchips; control presentation follows the approved product configuration |
| Pack sizes      | 48 and 96 tests where supported by a product-specific supplier IFU; other claims require confirmation  |
| Evidence status | Product-specific supplier IFU available.                                                               |

**Approval status:** Approved for external release as a controlled Safety Data Sheet. This document reflects the information available at the issue date; product-specific limitations stated in the document remain applicable and are managed through change control.

## 1. Identification

|                               |                                                                                                                                                                                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product identifier            | Mugen-Plex HIV-1 — Nucleic Acid Detection Kit (Lyophilized Fluorescent PCR Method)                                                                                                                                                                        |
| Recommended use               | Qualitative laboratory detection of human immunodeficiency virus type 1 (HIV-1) RNA in serum by one-step reverse-transcription real-time fluorescent PCR; for trained laboratory personnel.                                                               |
| Restrictions on use           | For research use only unless and until otherwise authorized. Not for self-testing. Not to be used alone to confirm or exclude infection.                                                                                                                  |
| Brand owner / local contact   | Precision Life Sciences (Private) Limited   Office #09, Police Colony, Nasir Bagh Road, Peshawar, Khyber Pakhtunkhwa, Pakistan   <a href="http://www.precisionlifesciences.com.pk/contact">www.precisionlifesciences.com.pk/contact</a>   +92 301 8182903 |
| Original reagent manufacturer | Shanghai Janzy Biotechnology Co., Ltd.   Supplier contact stated in IFU: Ms. Qiao, 18105338426                                                                                                                                                            |
| Emergency information         | No dedicated emergency number is provided. For exposure, contact local medical services and provide this SDS; for fire or spill incidents, follow the facility emergency procedure.                                                                       |

## 2. Hazard identification

|                          |                                                                                                                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GHS classification       | No GHS hazard classification is identified from the ingredient information currently available and the small lyophilized quantities described. Final classification remains dependent on complete quantitative formulation and raw-material supplier SDS review. |
| Signal word / pictograms | None assigned on the available information.                                                                                                                                                                                                                      |
| Other hazards            | Reconstituted or used tubes may contain human specimen nucleic acid and must be handled as potentially biohazardous. Lyophilized material and nucleic-acid controls are not identified as infectious virus. Avoid aerosol generation and contamination.          |

## 3. Composition / information on ingredients

|                     |                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detection microchip | Primers, dye-labelled hydrolysis probes, reverse transcriptase, Taq DNA polymerase, Tris, potassium chloride, magnesium sulfate and dNTPs. Quantitative concentrations and a complete excipient/preservative declaration were not supplied. |
| Controls            | Supplier IFU describes positive, negative and exogenous internal controls as plasmid material; no live virus or human-derived control material is identified.                                                                               |

|                              |                                                                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazardous ingredients</b> | None identified at reportable concentration from available information. Obtain and retain supplier SDS/CoA/origin declarations for all raw materials under the controlled material review process. |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4. First-aid measures

|                           |                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>         | Move to fresh air if dust/aerosol is inhaled. Seek medical advice if symptoms persist.                                                         |
| <b>Skin contact</b>       | Remove contaminated clothing. Wash skin with soap and water.                                                                                   |
| <b>Eye contact</b>        | Rinse cautiously with clean water for at least 15 minutes; remove contact lenses if easy to do. Seek medical advice for persistent irritation. |
| <b>Ingestion</b>          | Rinse mouth. Do not induce vomiting. Seek medical advice and show this SDS.                                                                    |
| <b>Notes to physician</b> | Treat symptomatically. No specific antidote is identified.                                                                                     |

## 5. Fire-fighting measures

|                             |                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable media</b>       | Use water spray, dry chemical, carbon dioxide or foam appropriate to the surrounding fire.                                                      |
| <b>Specific hazards</b>     | The supplied kit is not identified as flammable. Thermal decomposition of packaging or reagents may produce irritating fumes.                   |
| <b>Protective equipment</b> | Firefighters should use standard protective equipment and self-contained breathing apparatus where smoke or decomposition products are present. |

## 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b>      | Restrict access; wear laboratory coat, gloves and eye protection. Avoid raising dust or generating aerosols.                                                                                                                                       |
| <b>Cleanup</b>                   | For unopened/dry material, collect carefully into a closable container and wipe the area with water followed by an appropriate laboratory disinfectant. For used or specimen-contaminated material, follow the facility biohazard spill procedure. |
| <b>Environmental precautions</b> | Prevent uncontrolled release to drains or the environment.                                                                                                                                                                                         |

## 7. Handling and storage

|                               |                                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b>               | For trained laboratory personnel. Keep pre-amplification, specimen-processing and amplification areas separated. Do not mix components from different lots. Prevent moisture, aerosol and nucleic-acid contamination. |
| <b>Unopened storage</b>       | Product-specific supplier IFU: store and transport dry at room temperature; 24 months unopened.                                                                                                                       |
| <b>After opening</b>          | When resealed, up to 3 days; for longer storage after opening, freeze at $-20^{\circ}\text{C}$ .                                                                                                                      |
| <b>Reconstituted controls</b> | For products with a reviewed IFU: store at $-25$ to $-10^{\circ}\text{C}$ in aliquots and avoid repeated freeze–thaw cycles. For source-limited products, this claim remains unverified.                              |
| <b>Shelf life</b>             | 24 months unopened only where stated by a product-specific supplier IFU; final label claims are governed by approved stability evidence and applicable product labelling.                                             |

## 8. Exposure controls / personal protection

|                      |                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Occupational limits  | No specific occupational exposure limits identified for the disclosed mixture.                                                                                                                          |
| Engineering controls | Use normal molecular-laboratory ventilation and physical separation of clean, extraction and amplification areas. Use a biological safety cabinet when handling potentially infectious human specimens. |
| PPE                  | Laboratory coat, disposable gloves and safety glasses. Add face protection and other PPE according to specimen risk assessment and spill procedure.                                                     |
| Hygiene              | Do not eat, drink or apply cosmetics in the laboratory. Wash hands after handling and before leaving the laboratory.                                                                                    |

## 9. Physical and chemical properties

|                                                                                     |                                                                                                                                                    |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance                                                                          | Lyophilized microsphere/reagent in PCR tube-strip microchips. Bill of materials and control presentation follow the approved device master record. |
| Odour                                                                               | Not determined / expected to be minimal.                                                                                                           |
| pH, melting point, boiling point, vapour pressure, density, solubility, flash point | Not determined for the mixture.                                                                                                                    |

## 10. Stability and reactivity

|                         |                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Reactivity              | No hazardous reactivity identified under recommended handling.                                          |
| Stability               | Expected to be stable when kept dry and stored as specified in the product-specific supplier IFU.       |
| Conditions to avoid     | Moisture, excessive heat, direct sunlight, contamination, use after expiry and incompatible lot mixing. |
| Incompatible materials  | Strong oxidizing agents and conditions that denature enzymes or damage nucleic acids.                   |
| Hazardous decomposition | May emit irritating fumes if packaging or organic components burn.                                      |

## 11. Toxicological information

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|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Likely routes                                                    | Eye/skin contact, inhalation of dust/aerosol, accidental ingestion.                                                                                                       |
| Acute/chronic effects                                            | No mixture-specific toxicology studies or quantitative data were provided. No hazardous component is identified at a reportable concentration from available information. |
| Sensitization/carcinogenicity/mutagenicity/reproductive toxicity | No data available for the finished mixture. Enzyme preparations may cause sensitization in susceptible individuals after repeated occupational exposure.                  |

## 12. Ecological information

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|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Ecotoxicity, persistence, bioaccumulation and mobility | No finished-mixture data available.                                                                       |
| Precaution                                             | Avoid uncontrolled release. Dispose of used assay materials through the approved laboratory-waste stream. |

## 13. Disposal considerations

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| Unused product | Dispose according to institutional chemical/laboratory-waste procedures and applicable Pakistani requirements. |
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|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Used/specimen-contact materials</b> | Treat as potentially infectious laboratory waste. Segregate, decontaminate/autoclave where required and hand over through the licensed healthcare-waste route. Do not dispose of specimen-contact material as general waste. |
| <b>Packaging</b>                       | Uncontaminated packaging may follow approved recycling/general-waste procedures; contaminated packaging follows the biohazard stream.                                                                                        |

## 14. Transport information

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|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UN number / proper shipping name / class / packing group</b> | Not assigned from the available disclosed composition.                                                                                                                                                                                                         |
| <b>Transport status</b>                                         | The unused finished kit is not identified as dangerous goods on the available information. Final shipper classification must be confirmed after complete formulation and supplier SDS review and must account for any dry ice used for reconstituted controls. |

## 15. Regulatory information

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|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pakistan</b>                 | Product registration is governed by the DRAP Act, 2012 and Medical Devices Rules, 2017, as applicable. This SDS does not replace product registration, approved labelling, biosafety procedures or raw-material SDS. |
| <b>Workplace classification</b> | Final GHS classification and label elements require confirmation against the complete quantitative formulation and supplier component SDS.                                                                           |

## 16. Other information

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|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Document number</b>              | PLS-SDS-MP-HIV-001                                                                                                                                                                          |
| <b>Revision / issue date</b>        | Approved v1.0 / 09 October 2026                                                                                                                                                             |
| <b>Evidence status</b>              | Product-specific supplier IFU available.                                                                                                                                                    |
| <b>Sources</b>                      | HIV-1 English IFU, revised 15 August 2026; source SDS v0.1; KMU Quality Manual; KMU Waste Segregation Protocol; DRAP Medical Devices Rules, 2017.                                           |
| <b>Approval status</b>              | APPROVED FOR EXTERNAL RELEASE. Classification is based on information available at the issue date; the stated evidence limitations remain subject to documented change control.             |
| <b>Controlled information notes</b> | Resolve serum versus plasma and dry room-temperature versus 2–8 °C/-20 °C claims; confirm quantitative formulation, control identities, final kit configuration, REF and emergency contact. |